STRUCTURED QUESTIONS (100 MARKS)

(INSTRUCTION: Please answer ALL questions in the space provided.)

QUESTION 1 (15 Marks)

(a) Determine the sum generated by the 3-bit parallel adder in Figure 1 by completing the values in Table 1 for input A and B equal to 101_2 and 110_2 are being added. [4 marks]

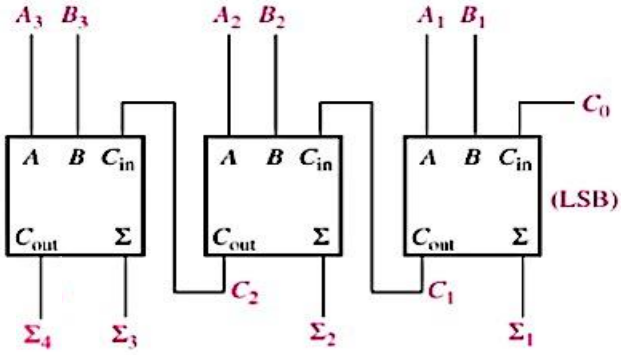


Figure 1

Table 1

Label	A_1	A_2	A_3	B_1	B_2	B_3	C_0	C_1	C_2	$\sum_1$	$\sum_2$	$\sum_3$	$\sum_4$
Value (binary)													

(b) Identify the comparator inputs and outputs when the binary numbers of A and B are applied as shown in Figure 2. Complete Table 2. [5 marks]

Table 2

Symbol	Value (Binary)
A	
B	
x	
y	
z	

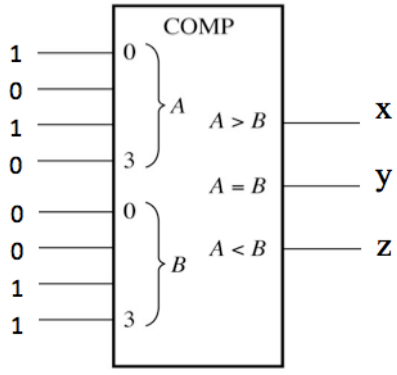


Figure 2

(c) Based on Figure 3, the data-input, D_i (i = 0, 1, 2, 3) and data-select, S_j (j = 0, 1) waveforms are applied to the multiplexer.

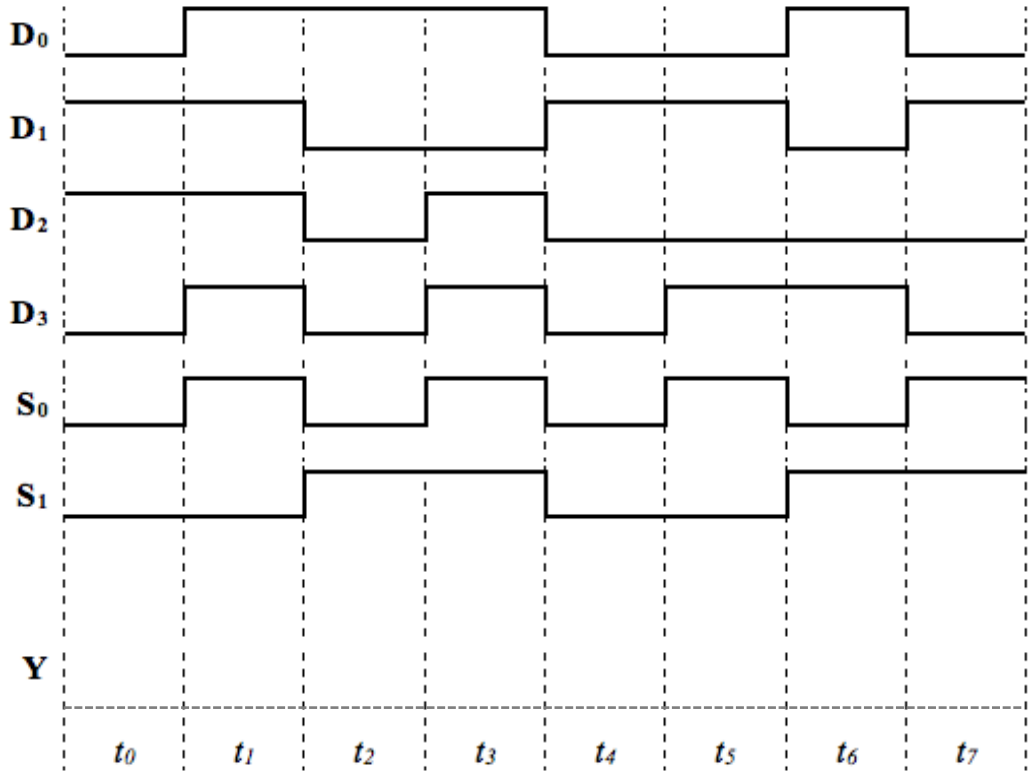


Figure 3

i. Identify the selection line (D_i) at t_0 until t_7 by filling in Table 3. [2 marks]

Table 3

Time	t_0	t_1	t_2	t_3	t_4	t_5	t_6	t_7
Selection line, D_i								

ii. Draw the output waveform Y in Figure 3. [4 marks]